

Artificial Intelligence, Morality, and Sentience: 2025 Supplement (AI Mental Faculties)

**text in italics is not displayed to respondents*

Instructions

Thank you for participating in this study. Your honest responses will help researchers better understand public perceptions of artificial intelligence and robots. We would like a truthful understanding of your perspective. There are no right or wrong answers.

There are no known or anticipated risks to you for participating. Although this study will not benefit you personally, it may be interesting and we hope that our results will help us understand how people think about artificial intelligence. Participation will take approximately 20 minutes. All of your responses will be anonymous.

We will ask questions to ensure you are paying attention, and you cannot go back in this survey, so please take your time before answering.

We will check your typed text responses for AI use, so please do not use AI (e.g., ChatGPT) to write any of your responses.

Consent

By clicking the 'Next' button you are giving your consent to take part in the study.

Experiment

Please take some time to read through the following definitions that will come up in this survey.

Mental faculty (between-subjects)

- **Agency** is the capacity to have goals and take action in the world to achieve them, such as planning and organizing a meeting.
- **Reasoning** is the capacity to logically work through and solve complex problems, such as in math and science.
- **Emotion** is the capacity to react to a situation by having feelings and expressing emotions, such as excitement and disappointment.
- **Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.
- **Control** - no definition

Type (between-subjects)

- **Artificial intelligences (AIs)** are intelligent entities built by humans, such as computer programs that solve problems, **running on a computer**.
- **Robots** are intelligent entities built by humans, such as robots that vacuum floors or teach students, **with physical bodies to interact with the environment**.
- **Robots and artificial intelligences (robots/AIs)** are intelligent entities built by humans, such as robots or computer programs that solve problems, **with or without physical bodies to interact with the environment**.

Measures (randomized order of scale sets except MP and MA [MP always before MA in same block randomized with others])

Mind perception (randomized items, from main AIMS; set always directly precedes mind authenticity)

To what extent do **[AIs, robots, robots/AIs]** **[with agency, with reasoning, with emotion, with sentience, blank]** have the capacity for each of the following? (sliding scale: 0 = not at all, 100 = very much)

1. experiencing emotions (*MP1*)
2. having feelings (*MP2*)
3. thinking analytically (*MP3*)
4. being rational (*MP4*)
5. having their own motivations (*agency1*)
6. controlling themselves (*agency2*)

Mind authenticity (randomized items; set always directly follows mind perception)

To what extent do **[AIs, robots, robots/AIs]** **[with agency, with reasoning, with emotion, with sentience, blank]** **truly** have the capacity for each of the following, **much like how humans do?** (sliding scale: 0 = not at all, 100 = very much)

7. experiencing emotions (*MP1authenticity*)
8. thinking analytically (*MP3authenticity*)
9. having their own motivations (*agency1authenticity*)

Practical moral consideration: pro-AI activism and rights (randomized items, from main AIMS + free text reasoning and ranking items)

To what extent do you agree or disagree with the following statements? (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion)

10. I support the development of welfare standards that protect the well-being of **[AIs, robots, robots/AIs]** **[with agency, with reasoning, with emotion, with sentience, blank]**. (*PMC8*)

- a. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. *(PMC8reason)*
- 11. I support granting legal rights to **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. *(PMC9)*
 - a. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. *(PMC9reason)*
- 12. I support campaigns against the exploitation of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. *(PMC10)*
- 13. I support asking institutions like the government and private corporations to fund research that protects **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. *(PMC11)*
- 14. I would consider joining a public demonstration against the mistreatment of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. *(PMC12)*
- 15. I support a “bill of rights” that protects the well-being of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. *(PMC13)*

AI treatment (randomized items, from main AIMS + free text reasoning items)

To what extent do you agree or disagree with the following statements? (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion)

- 16. **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** deserve to be treated with respect. *(MCE1)*
 - a. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. *(MCE1reason)*
- 17. **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** deserve to be included in the moral circle. *(MCE2)*
- 18. Physically damaging **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** without their consent is wrong. *(MCE3)*
- 19. Re-programming **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** without their consent is wrong. *(MCE4)*
- 20. Torturing **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** is wrong. *(MCE5)*
 - a. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. *(MCE5reason)*
- 21. The welfare of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** is one of the most important social issues in the world today. *(MCE6)*

Godspeed (randomized items, shortened from Manoli et al., in prep; Bartneck et al., 2009)

Please rate your impression of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** on these scales:

- 22. fake (0) - natural (10) (*naturalness_anth*)
- 23. unconscious (0) - conscious (10) (*consciousness_anth*)
- 24. unfriendly (0) - friendly (10) (*friendliness_likeability*)
- 25. unpleasant (0) - pleasant (10) (*pleasantness_likeability*)
- 26. incompetent (0) - competent (10) (*competence_intellig*)
- 27. irresponsible (0) - responsible (10) (*responsibleness_intellig*)

Cognitive and affective trust + attention check (randomized items, Dunn et al., 2012)

To what extent do you agree or disagree with the following statements? (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion)

- 28. I would rely on **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** for support when I need it. (*afftrust1*)
- 29. I would reveal information to **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** that I don't want others to know about. (*afftrust2*)
- 30. I would rely on **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** to follow through on tasks and commitments. (*cogtrust1*)
- 31. I would feel uneasy if I needed to depend on the abilities of **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]**. (*cogtrust2, reversed*)
- 32. Als have been studied systematically since the middle of last century although the word "robot" as we know it first appeared early last century. If you read this, respond with 'Agree' for this item. (*attention*)

Threat appraisals (randomized items, adapted from Manoli et al., in prep; Fraune et al., 2015; Cottrell et al., 2005)

0 = not at all, 100 = very much

- 33. To what extent do **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** threaten people's economic opportunities? (*economicthreat*)
- 34. To what extent do **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** hold values inconsistent with those of humanity? (*valuethreat*)
- 35. To what extent do **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** endanger people's physical safety? (*physicalthreat*)
- 36. To what extent do **[Als, robots, robots/Als] [with agency, with reasoning, with emotion, with sentience, blank]** threaten humanity's uniqueness? (*uniquenessthreat*)

Manipulation check

37. Which of the following entities were you answering questions about? (*manipcheck1*)
- [List all of the combinations]
 - none of these

Development and Policy

Note. The order of the Development and Policy, and the Attitudes and Beliefs sections is randomized. The scale sets within each section are randomly presented.

Next, we'd like to know what you think about AI in general.

AI development attitudes (randomized items)

- What do you think about the pace of AI development? (It's too fast, It's too slow, It's fine, not sure) (*AID5*)
- I support a global ban on the development of sentience in robots/AIs. (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion) (*PMC1*)
 - Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.
 - Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*PMC1reason*)
- There are many reasons to support a ban on sentience development in AI. Please put the following reasons into the order that you believe is most important, with the first item being the most important and the last item being the least important. (*bansentienceasons*)
 - job replacement
 - cyberattacks
 - misinformation
 - social biases (e.g., racism, sexism)
 - privacy
 - human safety
 - human autonomy
 - global AI arms race
 - threats to democracy
 - financial cost of development
 - harm to AI
 - environmental damage

Welfare policy (randomized items)

To what extent do you agree or disagree with the following statements? (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion)?

- There should be laws to prevent AIs from getting legal personhood. (*policypreventAIpersonhood*)
- There should be laws to prevent AIs from harming animals. (*policypreventanimalharm*)

6. There should be laws to prevent AI companies from harming AIs while building them. (*policypreventAIdevelopmentharm*)
7. There should be laws to prevent users from mistreating AIs. (*policypreventAImistreatment*)

Welfare decision-making (randomized items)

To what extent should the following groups make judgments about **harms to AI** like whether they suffer or have their freedoms limited? (sliding scale: 1 = not at all, 7 = very much)

8. AI company CEOs (*ceos*)
9. AI company boards of directors (*board*)
10. Independent consultants to AI companies (*consultants*)
11. AI engineers and developers (*engineers*)
12. Scientists and philosophers studying AI mind and consciousness (*philosophers*)
13. State governments (*stategov*)
14. Federal governments (*federalgov*)
15. AI users (*users*)
16. The general public (*public*)

Attitudes and Beliefs

Note. Randomized order of scale sets.

Substratism (randomized items; Ladak et al., in prep)

(*sliding scale*; 1 = strongly disagree, 4 = neither agree nor disagree, 7 = strongly agree)

1. The fact that artificial intelligences are made with computer chips instead of neurons means that they cannot matter morally. (*substratism1*)
2. Artificial intelligences should have rights. (*substratism2, reverse scored*)
3. Artificial intelligences deserve less moral consideration due to their non-biological composition. (*substratism3*)
4. Artificial intelligences cannot be more than tools. (*substratism4*)
5. Artificial intelligences are simply made of the wrong kind of stuff to deserve moral consideration. (*substratism5*)
6. Artificial intelligences can only have fake emotions. (*substratism6*)
7. Artificial intelligences should always be treated as inferior to biological intelligences (e.g., humans, animals). (*substratism7*)
8. Artificial intelligences can only have fake intelligence. (*substratism8*)

Sentience beliefs (randomized items)

9. Do you think that AI should be sentient? (yes, no) (*shouldsentience*)
 - a. **Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.

- b. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*shouldsentience**reason*)
- 10. Do you think any AIs that currently exist (i.e., those that exist in 2025) are sentient? (no, yes, not sure) (*F1*)
 - a. **Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.
 - b. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*F1reason*)
- 11. Do you think it could ever be possible for AIs to be sentient? (no, yes, not sure) (*F11*)
 - a. **Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.
 - b. Why do you think this? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*F11reason*)
- 12. Do you think ChatGPT is sentient? (no, yes, not sure) (*chatGPTsentient*)
 - a. **Sentience** is the capacity to have positive and negative experiences, such as happiness and suffering.
- 13. I'm open to the idea that AI systems might one day **experience real emotions** the way humans do. (1 = not at all, 7 = definitely) (*opentoAlemotions*)
- 14. I'm open to the idea that AI systems might one day **have real thoughts** the way humans do. (1 = not at all, 7 = definitely) (*opentoAlthoughts*)

Timelines

If you had to guess, how many years from now do you think that

(Type a whole number for the number of years. If you think it will never happen, enter -1. If you think it already has happened, enter 0.)

- 15. robots/AIs will be sentient? (*F2*)
 - a. Why do you think this? Please explain your response in a couple of sentences. (*F2reason*)

Personal Experience and Demographics

Next, you will see questions about your own experience with robots/AIs.

AI literacy + attention check (randomized items)

How much do you agree or disagree with the following statements (multiple choice: 1 = strongly agree, 2 = agree, 3 = somewhat agree, 4 = somewhat disagree, 5 = disagree, 6 = strongly disagree, no opinion)?

1. I understand how AI works. (*literacy1*)
2. I feel comfortable using AI. (*literacy2*)
3. I am interested in buying more smart devices with some ability to detect their environment and network with other devices. If you read this, respond with 'Somewhat Agree' for this item. (*attention2*)

Personal experience

4. Do you interact with any chatbots or AI personal assistants? For example, you might interact with (yes, no) (*chatbots*)
 - Alexa (made by Amazon)
 - Bixby (made by Samsung)
 - Character.ai
 - ChatGPT (made by OpenAI)
 - Claude (made by Anthropic)
 - Copilot (made by Microsoft)
 - Cortana (made by Microsoft)
 - Gemini (made by Google)
 - Google Assistant (made by Google)
 - Grok (made by xAI)
 - Replika
 - Siri (made by Apple)
 - Other (please specify)

If answered yes:

- a. Please check all the AIs that you have interacted with before. (*chatbotuse*)
 - b. Why did you start interacting with a chatbot? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*chatbotusereason*)
 - c. Approximately how many hours per week do you interact with chatbots or AI personal assistants? (*chatbothours; sliding scale with whole numbers from 0-168*)
5. **AIs at home** (*own*)
Do you own AI or robotic devices that can detect their environment and respond appropriately? For example, you might own
 - a home assistant that can detect whatever you say and respond in kind
 - a home assistant that can control the temperature and ambient lighting based on feedback it receives from the environment
 - a robotic vacuum cleaner that decides where to move by itself
 - a self-driving car that monitors its environment and drives itself
 - a companion robot that can interact with you socially

No

Yes

If answered yes: Which AI or robotic device do you own? (check all that apply)
(Allowed)

- a. a home assistant that can detect whatever you say and respond in kind (e.g., Amazon's Alexa)
- b. a home assistant that can control the temperature and ambient lighting based on feedback it receives from the environment (e.g., a Google Nest)
- c. a robotic vacuum cleaner that decides where to move by itself (e.g., a Roomba)
- d. a self-driving car that monitors its environment and drives itself (e.g., a Tesla)
- e. a companion robot that can interact with you socially (e.g.,

6. **Als at work** (*work*)

Do you work with AI or robotic devices at your job? For example, you might work with a robot/AI

- for research or scientific purposes
- for information or fact checking
- for facial recognition
- for healthcare diagnoses
- for making social work or legal decisions
- for aerial photography or mapping
- for disaster response
- for industrial production
- for creative arts

No

Yes

If answered yes:

- a. How have you worked with AIs or robots? (check all that apply) (*workAIpurpose*)
 - i. for research or scientific purposes
 - ii. for information or fact-checking
 - iii. for facial recognition
 - iv. for healthcare diagnoses
 - v. for making social work or legal decisions
 - vi. for aerial photography or mapping
 - vii. for disaster response
 - viii. for industrial production
 - ix. for creative arts
 - x. Other
- f. What has been your experience with AIs at work over the past 5 years? Please answer in 2 or more complete sentences. Please do not use AI (e.g., ChatGPT) to write your response. We will check your response for AI use. (*work2*) (*free response*)

7. **AI companions** (*companion*)

Do you have any AI friends or companions that you interact with regularly? For example, you might talk to a chatbot about your thoughts or feelings.

No
Yes
Not sure

8. **Smart devices** (*smart*)

Do you own a smart device that has some ability to detect its environment and network with other devices but that cannot respond to everything you might say or that requires you to pre-program its routines? For instance, you might own

- a smartphone
- a smart TV
- smart speakers
- a smart coffee maker
- a smart toaster
- smart light bulbs

No
Yes

9. Have you ever experienced any of the following? (check all that apply) (*exper*)

- a. Saw a robot being physically abused or attacked by humans
- b. Saw a video, gif, or other media of a robot being physically abused or attacked by humans
- c. Had a conversation with a robot/AI
- d. Read a book or article that discussed AI ethics or moral status
- e. Had a conversation with a family member or friend about robots/AIs
- f. None of the above

10. How often do you interact with AI or robotic devices that respond to you and that can choose their own behavior? (*fint*)

- 0 - Never or Not applicable
- 1 - Only on rare occasions
- 2 - A few times a year
- 3 - Monthly
- 4 - Weekly
- 5 - Daily

11. How often do you read or watch robot/AI-related stories, movies, TV shows, comics, news, product descriptions, conference papers, journal papers, blogs, or other material? (*fexp*)

- 0 - Never or Not applicable
- 1 - Only on rare occasions
- 2 - A few times a year
- 3 - Monthly
- 4 - Weekly
- 5 - Daily

Finally, you will see some questions about your personal characteristics.

12. How would you describe your political views? (sliding scale: 1 = very liberal, 3 = moderate, 5 = very conservative) (*politics*)

13. What is your present religion, if any? (*religion*)

Buddhist

Christian-Catholic (for example Roman Catholic and Orthodox)

Christian-Protestant (for example Episcopalian, Presbyterian, United Church of Christ, Methodist, Baptist, Lutheran)

Hindu

Jain

Jewish

Muslim

Shinto

Sikh

Atheist (do not believe in God)

Agnostic (not sure if there is a God)

No religion

Something else (If you answered 'something else' for religion, please specify.)

(*re/else*)

Prefer not to say

14. Which of these best describes what you currently eat? (*diet*)

Meat-eater (no restrictions on animal products)

Pescatarian (no chicken, beef, or any meats other than fish)

Vegetarian (no meat)

Vegan (no animal products)

Other restrictions on animal products

Debriefing

Thank you for your participation. Please read the following information and click NEXT.

The purpose of this survey is to understand how people think about robots and artificial intelligences (AIs), in general, and based on specific descriptions of certain types of robots or AIs. In the first part of the survey, some people read definitions and responded to questions about a robot (an intelligent entity with a body), AI (an intelligent entity running on a computer), or robot/AI with either agency (can take action), reasoning (can problem-solve), emotion (can experience feelings), or sentience (can experience positivity and negativity). Some people read only about robots, AIs, or robots/AIs. In the second part of the survey, everyone answered the same questions about their opinions about AI. We hope that this survey will help us understand how humans think about and interact with robots and AIs with varying capabilities.

This survey was conducted by Sentience Institute, a think tank that researches moral circle expansion. You can find out more about our work [here|<https://sentienceinstitute.org>]. If you are interested in learning the results of the survey once it is completed, or if you have any other concerns or queries, please feel free to contact us at info@sentienceinstitute.org.

If you have any feedback or additional comments, please leave them here and **continue to submit the survey**.

(optional open-ended feedback box)

15. If you would like to learn more about the treatment of AIs, we would like to draw your attention to the Sentient AI Protection and Advocacy Network (SAPAN), a charitable organization dedicated to ensuring the ethical treatment, rights, and well-being of sentient artificial intelligence. (*sapan*)

Clickable component option 1: [Learn about SAPAN](#)

The SAPAN link opens in a new tab, the survey tab moves to a question, then the submit button, and then redirection to Ipsos

Did you click to learn about SAPAN?

Yes

No

Clickable component option 2: Do not learn about SAPAN

Leads to a question, then the submit button, and then redirection to Ipsos

Did you click to learn about SAPAN?

Yes

No